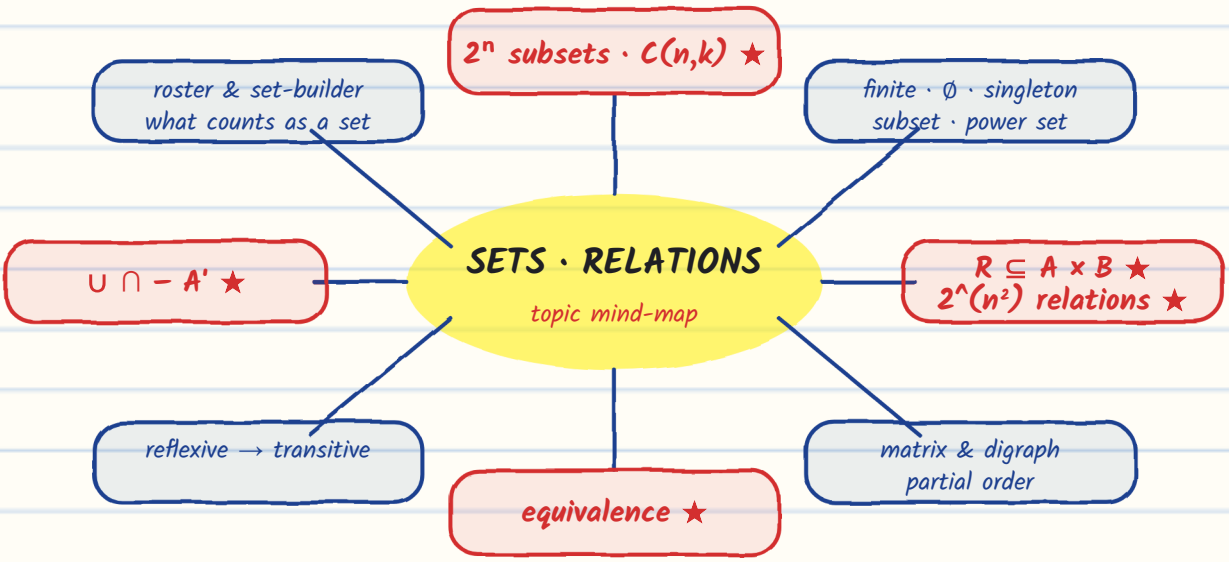
Set Theory, Relations & Functions

2^n subsets · set laws · relations $\subseteq A \times B$ · injective / onto: exam-ready in 11 sheets



↑ the topic on one page: red = high-yield, blue = definitions

➜ Syllabus coverage

"Discrete Mathematics, Theory of Computation and Basic Graph Theory: Set theory; Relations; Functions; Graph theory; Propositional calculus and predicate calculus; ..." This notebook covers the first three; graph theory and predicate calculus have their own lessons in the chapter.

High-yield list

① 4 elements ⇒ 2^4 = **16** subsets
② 2^n subsets; $C(n,k)$ of size k
③ $|A \cup B| = |A|+|B|-|A \cap B|$
④ De Morgan: $(A \cup B)' = A' \cap B'$
⑤ a relation = **subset** of $A \times B$
⑥ 3 elements ⇒ 2^9 = **512** relations
⑦ injective ⇒ $|A| \le |B|$; onto ⇒ $\ge$
⑧ $A \to B$ has $|B|^{|A|}$ functions
⑨ $(f \circ g)(x) = f(g(x))$: f **after** g
⑩ bijective ⇔ f^{-1} exists

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ What a set is; the two forms ... p. 2
✓ Counting: 2^n and $C(n,k)$ ... p. 2
✓ Types of sets & power set ... p. 3
✓ Set operations & the laws ... p. 4
✓ Relations $\subseteq A \times B$ ... p. 5
✓ Six properties, counter-examples ... p. 6
✓ Equivalence & partial order ... p. 7
✓ Functions & one-to-one / onto ... p. 8
✓ Composition, inverse, counting ... p. 9
✓ Night-before p. 10 · Practice p. 11

1 What Is a Set? definition ★

Definition A **set** is a **well-defined collection of distinct objects**, its **elements** or **members**. "Well-defined" means we can decide unambiguously **yes or no** whether an object belongs. The objects may be numbers, letters, points, or even other sets.

Form	It does what	Example
Roster (tabular)	lists **every** element one by one	$A = \{2, 4, 6, 8\}$
Set-builder	states the **property** the elements share	$A = \{x \mid x \text{ even}, x < 10\}$

Order and repetition are irrelevant $\{1, 2, 3\} = \{3, 2, 1\} = \{1, 1, 2, 2\}$ · but $(1, 2) \neq (2, 1)$ — **ordered pairs do care**
$|A|$ is read "the **cardinality** of A": $|\{a, b\}| = 2$.

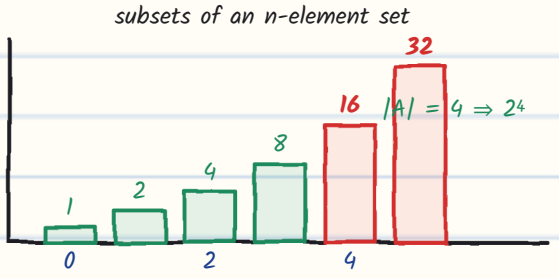
Exam trap {even numbers} is **not** a set — it is not well defined until you say which evens. That clause is the whole definition and it is asked often.

Memory trick A set is a **basket with a label**: the label decides who is in, and the order of picking never changes the contents.

2 Counting Subsets: 2^n and $C(n,k)$ most tested ★

The counting argument Walk through the **n** elements one at a time; each has exactly two choices — put it in, or leave it out. Two choices, n times, gives 2^n subsets, counting both **∅** and **A itself**.

n	$\lvert P(A)\rvert = 2^n$	of size 2	proper
0 ($A = \emptyset$)	1	0	0
2	4	1	3
3	8	3	7
4	**16**	**6**	**15**
5	32	10	31



every extra element **doubles** the count — 2^n grows faster than anything else here

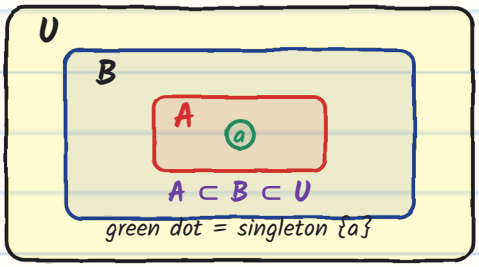
Worked example ✍ "A set A has **4** elements. How many subsets?" → **16**, not 8 and not 4. **Proper** subsets → **15**; excluding the empty set too → **14**. Subsets of size k = $C(n,k) = n!/k!(n-k)!$; for n = 4: 4 + 6 + 4 + 1 = 15, and adding ∅ gives 16 ✓

3 Types of Sets, Subsets & the Power Set

definition bank

Name	Definition	Example
Finite / infinite	a definite countable number of elements, $\|A\| = n$ / elements never run out	$\{1, 2, 3\}$ / $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
Null / empty set $\emptyset$	**no** elements at all; also written $\{\}$ or φ	$\emptyset = \{x \mid 1 = 0\}$
Singleton set	**exactly one** element	$\{a\}$
Equal sets	the same elements, in any order: $A = B$	$\{1, 2, 3\} = \{3, 2, 1\}$
Subset $A \subseteq B$	every element of A is in B	$\{1, 2\} \subseteq \{1, 2, 3\}$
Proper subset $A \subset B$	$A \subseteq B$ **and** $A \neq B$	$\{1, 2\} \subset \{1, 2, 3\}$
Universal set U	every element under discussion belongs to it	$U = \{1, 2, 3, 4, 5, 6\}$
Complement A'	everything in U that is **not** in A: $A' = U - A$	$U = \{1,2,3,4\}$, $A = \{1,2\} \Rightarrow \{3,4\}$

Power set $P(A) = \{ X \mid X \subseteq A \}$ — the set of **all** subsets of A. Then $|P(A)| = 2^{|A|}$ and, for every finite A, $|P(A)| > |A|$ (Cantor). Edge cases: $|P(\emptyset)| = 1$ and $P(\{a\}) = \{ \emptyset, \{a\} \}$, so $P(A)$ itself is a set and $P(P(A))$ exists too.



Memory trick $\subseteq$ is "all in", $\subset$ is "all in and more outside". Every set is a subset of itself ($A \subseteq A$) and $\emptyset$ is a subset of every set ($\emptyset \subseteq A$): $\emptyset$ is the smallest subset, U the largest.

nested the other way round reads as **superset**: $B \supset A$ means B has everything A has, and more

Worked example ✍ subsets of $A = \{1, 2, 3, 4\}$

total 16 · exactly 2 elements → 6 · exactly 3 → 4 · proper → 15 · proper and non-empty → 14. Since $|P(A)| = 16 > 4$, $A \subset P(A)$ strictly, and the largest subset is A itself.

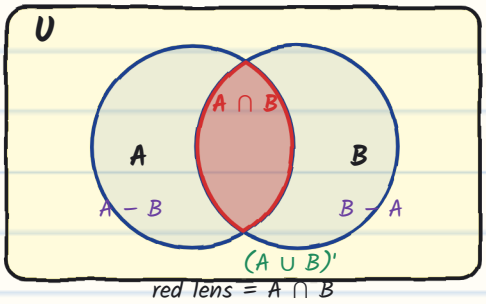
Exam trap (i) $\emptyset \subset \{x\}$ holds, but $\{x\} \subset \emptyset$ is **false** — $\emptyset$ has no elements to miss. (ii) The empty set is **not** equal to $\{0\}$ or $\{\emptyset\}$. (iii) "Subset" never means "proper subset" in ordinary exam wording.

IT link The power set is not an exam abstraction: an **n-bit flag word has exactly 2^n settings**, and asking "which of these 32 permissions is on?" is a power-set question in disguise. One bit per permission, every combination legal, nothing missing. The same count gives the number of **subsets a table can filter on**, which is why a query with n boolean predicates has at most 2^n distinct result shapes.

4 Set Operations and the Laws of Set Algebra high-yield ★

Operation	Symbol & definition	Worked instance
Union	$A \cup B = \{x \mid x \in A$ **or** $x \in B\}$ — everything in either	$\{1,2\} \cup \{2,3\} = \{1,2,3\}$
Intersection	$A \cap B = \{x \mid x \in A$ **and** $x \in B\}$ — the common part	$\{1,2\} \cap \{2,3\} = \{2\}$
Difference	$A - B = \{x \mid x \in A$ and $x \notin B\}$ — A with B removed	$\{1,2,3\} - \{2,3\} = \{1\}$
Complement	$A' = \{x \mid x \in U$ and $x \notin A\}$ — $U - A$	$U = \{1,2,3,4\}$, $A = \{1,2\} \Rightarrow \{3,4\}$
Cartesian product	$A \times B = \{(a, b) \mid a \in A, b \in B\}$ — all ordered pairs	$\{1,2\} \times \{x,y\} = \{(1,x),(1,y),(2,x),(2,y)\}$

Law	Statement
Commutative	$A \cup B = B \cup A$ · $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ · same for $\cap$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A$ · $A \cap U = A$
Complement	$A \cup A' = U$ · $A \cap A' = \emptyset$
Idempotent	$A \cup A = A$ · $A \cap A = A$
Involution	$(A')' = A$
Absorption	$A \cup (A \cap B) = A$ · $A \cap (A \cup B) = A$
De Morgan	$(A \cup B)' = A' \cap B'$ · $(A \cap B)' = A' \cup B'$



red lens = $A \cap B$

only the overlap is the intersection; everything inside **both circles** is the union

Cardinality formulas you must know

$|A \cup B| = |A| + |B| - |A \cap B|$

If $A \cap B = \emptyset$ (disjoint) this collapses to $|A| + |B|$. Generalising to three sets gives inclusion–exclusion, which is a separate lesson.

Exam trap $A - B$ and $B - A$ are **different sets**, so difference is **not** commutative: $\{1,2,3\} - \{2,3\} = \{1\}$ but $\{2,3\} - \{1,2,3\} = \emptyset$. And $A \times B \neq B \times A$ as sets, though $|A \times B| = |B \times A|$.

Worked example ✍ simplify with the laws $(A \cup B)' = A' \cap B'$ by De Morgan, and $((A \cup B)')' = A' \cap B'$ returns to $A \cup B$ by involution. Also $A - B = A \cap B'$, so $\{1,2,3\} - \{2,3\} = \{1,2,3\} \cap \{1\} = \{1\}$. De Morgan · involution · difference in terms of intersection and complement

5 A Relation Is a Subset of A × B

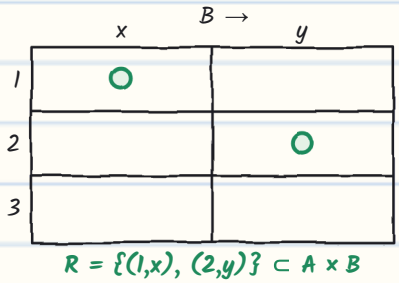
definition ★

Definition Let A and B be two sets. A relation R from A to B is any subset of $A \times B$, written $R \subseteq A \times B$. Its elements are **ordered pairs** (a, b) with $a \in A$, $b \in B$, read "a is related to b". If $R \subseteq A \times A$ it is a **binary relation on A**; that is the case whenever both sets are the same.

The two counting results $|A \times B| = |A| \cdot |B|$ — the pairs form a full grid, m rows by n columns.

Relations $A \to B = 2^{(|A| \cdot |B|)}$, because a relation is **any** subset and a set of $m \cdot n$ elements has $2^{(m \cdot n)}$ subsets.

On a single set with n elements: $|A \times A| = n^2$, so $2^{(n^2)}$ **binary relations**.



$R = \{(1,x), (2,y)\} \subset A \times B$

$A = \{1,2,3\}$, $B = \{x,y\}$: the grid has $|A| \cdot |B| = 6$ cells and a relation may mark **any** subset — $2^6 = 64$ relations $A \to B$

Worked example ✍ the counts that always appear

(a) $|A| = 3$: $|A \times A| = 9$, so $2^9 = 512$ relations on A. **(b)** $|A| = 3$, $|B| = 4$: $|A \times B| = 12$, so $2^{12} = 4096$ relations $A \to B$. **(c)** $|A| = 4$: $2^{16} = 65\,536$. The count explodes because the exponent is n^2, not n.

Exam trap **Not every relation is a function.** A relation may contain $(1, x)$ **and** $(1, y)$ for the same first element; a function may leave part of the codomain untouched. And $R \subseteq A \times B$ does **not** mean $R = A \times B$ — the universal relation is only one of the $2^{(m \cdot n)}$ choices.

IT link A database join condition is a **relation** between two tables, and a foreign key declares one.

→ Three ways to write the same relation

1. **Set of ordered pairs** — $R = \{(1, x), (2, y)\}$
2. **Matrix** $M_R = [m_{ij}]$, rows = A, columns = B, $m_{ij} = 1$ if $(a_i, b_j) \in R$, else 0
3. **Digraph** — one vertex per element of A and one per element of B, with a directed edge $a_i \to b_j$ for every pair in R. For a relation **on** A there is a single vertex set, and (a, a) is drawn as a self-loop

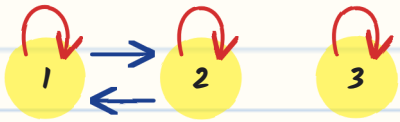
6 The Six Properties, with a Counter-example for Each

most tested ★

Property	Exact statement (on a set A)	Counter-example on A = {1, 2, 3}
Reflexive	$(a, a) \in R$ for **every** $a \in A$	$R = \{(1,2),(2,3)\}$ — $(3,3)$ is missing
Not reflexive (irreflexive)	$(a, a) \notin R$ for every $a \in A$	$R = \{(1,1)\}$ on $\{1,2,3\}$ — $(2,2)$ is present
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	$R = \{(1,2),(1,1)\}$ — $(2,1)$ is missing
Antisymmetric	$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$	$R = \{(1,2),(2,1)\}$ — both ways, yet $1 \neq 2$
Transitive	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$	$R = \{(1,2),(2,3)\}$ — forces $(1,3)$, missing
Asymmetric	$(a, b) \in R$ and $a \neq b \Rightarrow (b, a) \notin R$	$R = \{(1,2),(2,1)\}$ — symmetry holds instead

The distinction that loses marks

Symmetric and antisymmetric are not opposites and neither implies the other. $R = \{(1,2),(2,1)\}$ fails antisymmetry but **is** symmetric. $R = \{(1,2)\}$ is antisymmetric yet **not** symmetric. Only "asymmetric" is the true opposite of "symmetric".



$R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$

loops everywhere + $1 \leftrightarrow 2 \Rightarrow$ equivalence

reflexivity = a **loop at every** vertex; transitivity = every path of length two has a shortcut arrow

Memory trick **Symmetric = the arrow reverses; antisymmetric = at most one arrow per unordered pair.** Two more worth memorising: $\emptyset$ on a non-empty A is symmetric, antisymmetric and transitive but **not** reflexive, while $A \times A$ is reflexive, symmetric and transitive but **not** antisymmetric. The identity relation $\{(a,a)\}$ is the only one that passes all four.

Worked example ✍ matrix form For $R = \{(1,x), (2,y)\}$ with rows 1,2,3 and columns x,y: $M_R = [[1,0],[0,1],[0,0]]$. On a relation **on** A rows and columns are the same list, so **the diagonal is the reflexivity test.**

7 Equivalence Relations and Partial Orders

written favourite ★

Equivalence relation reflexive + **symmetric + transitive**, nothing more. Its classes $[a] = \{ x \in A \mid (a, x) \in R \}$ are non-empty, pairwise disjoint, and their union is A.

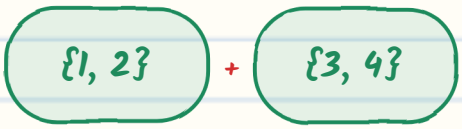
Equivalence relation ↔ partition, one-to-one.

Worked example ✍ $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ on $\{1,2,3\}$

Reflexive: three loops present. Symmetric: $(1,2)$ and $(2,1)$ both present. Transitive: $(1,2),(2,1)$ forces $(1,1)$ ✓ and $(2,1),(1,2)$ forces $(2,2)$ ✓. So R is an **equivalence relation** with classes $\{1, 2\}$ and $\{3\}$ — and it is **not** a partial order, since $(1,2)$ and $(2,1)$ with $1 \neq 2$ break antisymmetry.

Equivalence relations on a small set = Bell numbers

They correspond one-to-one to the partitions, so the count is B_n: $B_1 = 1 \cdot B_2 = 2 \cdot B_3 = 5 \cdot B_4 = 15 \cdot B_5 = 52$. A 4-element set gives $15 = 1$ (one block) $+ 7$ (two) $+ 6$ (three) $+ 1$ (four).



a partition of a 4-element set
⇒ one equivalence relation

classes never overlap and never leave a gap — that is exactly what "equivalence" means

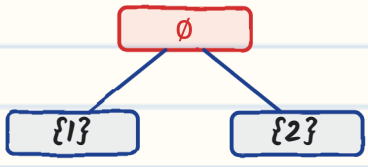
Partial order (poset) reflexive + **antisymmetric + transitive**. Add "every two elements are comparable" and you get a **total order**. The Hasse diagram drops the loops and keeps only the **covering** arrows upward.

Worked example ✍ congruence mod 3 on $\mathbb{Z}$

$a \, R \, b$ iff 3 divides $(a - b)$. Reflexive: $a - a = 0 = 3 \cdot 0$. Symmetric: $a - b = 3m \Rightarrow b - a = 3(-m)$. Transitive: $3m + 3k = 3(m + k)$. Classes $[0] = \{..., -6, 0, 3, 6, ...\}$, $[1] = \{..., -2, 1, 4, 7, ...\}$, $[2] = \{..., -1, 2, 5, 8, ...\}$ — disjoint, union $\mathbb{Z}$.

Memory trick Both are named for what they do: **E**quivalence lumps equal-looking things into **classes**; a partial order gives **P**artial **O**rdering, where some things are ranked and others are not comparable. $\subseteq$ **on $P(A)$ is always a partial order** — the standard example.

Exam trap A relation is **both** an equivalence relation and a partial order only when every class is a singleton, i.e. R is the identity relation. And "not symmetric" does **not** make a relation antisymmetric — antisymmetry is a separate test, not a negation.



$P(\{1,2\})$: a poset

loops and the arrows $\emptyset \longrightarrow \{1,2\}$ are omitted; $\{1\}$ and $\{2\}$ are **not comparable**

8 Functions: Domain, Codomain, Range

definition ★

Definition A function $f: A \to B$ is a relation from A to B in which every element of A appears exactly once as the first component of a pair. Equivalently, $(a_1, b_1), (a_2, b_2) \in R$ with $a_1 = a_2$ force $b_1 = b_2$. Nothing is said about how often an element of B is hit.

Name	It is	In code
Domain	the whole set A the function is declared on	every x that must be handled
Codomain	the declared target set B	the set the signature promises
Range / *image*	$f(A) = \{ f(a) \mid a \in A \} \subseteq B$ — only the values actually taken	the values you can really observe
Constant *function*	one fixed value for every input; the range is a singleton	the same constant returned always
Total / partial	total = every domain element has an image; partial may not	no null

→ Injective, surjective, bijective

Name	Test	Plain words
Injective *one-to-one*	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$	no two inputs share an output
Surjective *onto*	$\forall b \in B \; \exists a \in A : f(a) = b$	no output is left unused
Bijective	both of the above	a perfect pairing



injective, NOT onto



onto, NOT injective

bijective needs both: $|A| = |B|$

one green arrow per input — no output is hit twice, and none is skipped

The counting facts for finite sets

$|A| \le |B|$ is necessary for an injective $A \to B$

$|A| \ge |B|$ is necessary for a surjective $A \to B$

A bijection needs $|A| = |B|$, and then range = codomain. So if $|A| > |B|$ a function still exists but it **cannot** be injective; if $|A| < |B|$ it **cannot** be surjective.

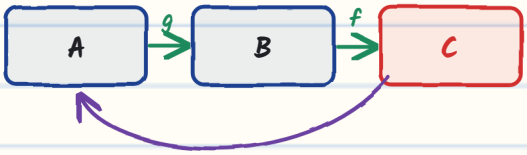
Exam trap **Injective and surjective are independent.** $f: \{1,2,3\} \to \{x,y\}$ must be onto but can never be one-to-one. A **constant** function is onto only when $|B| = 1$, and one-to-one only when $|A| = 1$.

9 Composition, Inverse, Identity — and Counting Functions

high-yield ★

Composition Let $g: A \to B$ and $f: B \to C$. Then $(f \circ g)(x) = f(g(x))$, a function $A \to C$. Read right to left: **g acts first, f afterwards.** The domain of $f \circ g$ is the domain of g. Composition is **associative** but **not commutative**.

$$(f \circ g)(x) = f(g(x))$$



the purple arc is f^{-1} running back $C \to B$

g first (green), then f — **read the composition right to left**

Inverse and identity The **identity** map $I_A(x) = x$ changes nothing. $f: A \to B$ is invertible ⇔ bijective; then $f^{-1}: B \to A$ exists and $f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$. Formally $f^{-1}(y) = x$ whenever $f(x) = y$.

Memory trick **$\circ$ is a "then" sign:** **$f \circ g$ means "f after g"**, exactly as the brackets $f(g(x))$ say. The types forbid $f \circ g$ unless the codomain of g fits the domain of f.

Counting functions $A \to B$ with $|A| = m$, $|B| = n$

All functions: n^m	Injective: $n(n-1)\cdots(n-m+1) = n!/(n-m)!$
Surjective: $n! \cdot S(m, n)$, S = Stirling	Bijections: $n!$ (only when $m = n$)

Exam trap (i) $(f \circ g)(x)$ is **$f(g(x))$**, not $g(f(x))$ — check with a concrete x. (ii) Composition of bijections is bijective and injective ∘ injective is injective, but the mixed cases are not guaranteed. (iii) A relation has an inverse function only if it is a **bijection**, not merely injective.

Worked example ✍ $|A| = 3$, $|B| = 4$ Each of the 3 inputs picks any of 4 outputs independently → $4^3 = 64$ functions. ($24 = 4 \cdot 3 \cdot 2$ counts only the injective ones; $81 = 3^4$ would be functions $B \to A$.) Surjective: 0, because 3 inputs cannot cover 4 outputs. Bijections: 0.

Worked example ✍ inverse and composition

$f(x) = 2x + 3$. Let $y = 2x + 3 \Rightarrow x = (y - 3)/2$, so $f^{-1}(x) = (x - 3)/2$, and $f((x - 3)/2) = x$ checks it out. Note $1/(2x + 3)$ is the reciprocal, never the inverse. For $f(x) = 2x + 1$ and $g(x) = x^2$: $(f \circ g)(x) = 2x^2 + 1$ but $(g \circ f)(x) = (2x + 1)^2$ — **not equal**.

Night-Before-the-Exam Sheet

Fact sheet

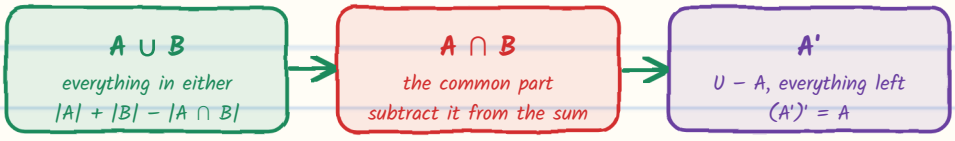
$\|A\| = n \Rightarrow \|P(A)\| = 2^n$ subsets	of size k: $C(n,k) = n!/k!(n-k)!$
$\|A \cup B\| = \|A\| + \|B\| - \|A \cap B\|$	disjoint $\Rightarrow \|A \cup B\| = \|A\| + \|B\|$
$A \cup A' = U \cdot A \cap A' = \emptyset$	$(A')' = A \cdot A - B = A \cap B'$
De Morgan: $(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$
$\|A \times B\| = \|A\| \cdot \|B\|$	relations $A \to B = 2^{(\|A\|\|B\|)}$
on an n-set: $2^{(n^2)}$ relations	$n = 3 \Rightarrow$ **512**
injective $\Rightarrow \|A\| \leq \|B\|$	surjective $\Rightarrow \|A\| \geq \|B\|$
bijection $\Rightarrow \|A\| = \|B\| \Leftrightarrow f^{-1}$ exists	$A \to B$: $\|B\|^{\|A\|}$ functions
$(f \circ g)(x) = f(g(x))$ — f after g	equivalence = R + S + T · poset = R + anti + T
bijections on an n-set: $n!$	injective count: $n!/(n-m)!$

→ Top 12 one-liners for MCQs

1. 4 elements ⇒ **16** subsets, 15 proper
2. Sets ignore order and repeats
3. $|P(\emptyset)| = 1$, $|P(\{a\})| = 2$
4. $|A \cup B| = |A|+|B|-|A \cap B|$
5. $A \times B \neq B \times A$
6. 3 elements ⇒ $2^9 = 512$ relations
7. Relation = **any** subset of $A \times B$
8. Symmetric ≠ antisymmetric
9. Empty relation: symmetric, anti, transitive
10. $A \to B = |B|^{|A|}$ functions
11. f^{-1} of $2x+3$ is $(x-3)/2$
12. $f \circ g$ means **f after g**

Top traps counting subsets as $2^n - 1$ without reading "proper" · treating **symmetric** and **antisymmetric** as opposites · writing 2^n relations on an n-set when it is $2^{(n^2)}$ · "every relation is a function" · "injective implies surjective" · swapping $f \circ g$ into $g(f(x))$ · calling $1/(2x+3)$ an inverse

All tricks together "A set is a basket with a label" · "in or out, n times, gives 2^n" · "$\subseteq$ is all-in, $\subset$ is all-in and more outside" · "a relation is any subset of the grid" · "symmetric reverses the arrow, antisymmetric allows at most one" · "equivalence lumps equal things into classes" · "$\circ$ is a then-sign: f after g" · "bijective is the only kind that can be undone"



the three operations the whole MCQ paper is built on

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. A set A has 4 elements. How many subsets?
(a) 8 (b) 16
(c) 4 (d) 32

2. $A = \{1,2,3\}$, $B = \{2,3,4\}$. $A \cap B$ is:
(a) $\{1,4\}$ (b) $\{1,2,3,4\}$
(c) $\{2,3\}$ (d) $\{1\}$

3. Relations on a 3-element set:
(a) 9 (b) 64
(c) 27 (d) 512

4. $|A| = 3$, $|B| = 4$. Functions $A \to B$:
(a) 12 (b) 64
(c) 81 (d) 24

5. $f(x) = 2x + 3$. Its inverse is:
(a) $(x+3)/2$ (b) $2x - 3$
(c) $(x-3)/2$ (d) $1/(2x+3)$

6. Equivalence relations on a 4-element set:
(a) 16 (b) 52
(c) 15 (d) 24

7. $\{(1,2),(2,1)\}$ on $\{1,2,3\}$ is:
(a) symmetric only (b) antisymmetric
(c) both (d) neither

8. $(A \cup B)'$ equals:
(a) $A' \cup B'$ (b) $A' \cap B'$
(c) $A \cap B$ (d) $A - B$

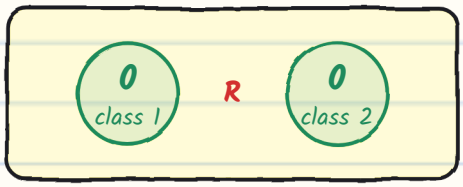
9. $|A| = 5$, $|B| = 2$. A function $A \to B$ must be:
(a) surjective (b) injective
(c) bijective (d) constant

10. Subsets of $A = \{1,2,3\}$ of size 2:
(a) 2 (b) 3
(c) 6 (d) 8

Answer key → 1-b · 2-c · 3-d · 4-b · 5-c · 6-c · 7-a · 8-b · 9-a · 10-b

Written question On the integers $\mathbb{Z}$ define $a \; R \; b$ iff 3 divides $(a - b)$. Prove R is an equivalence relation, list its classes, then give on $A = \{1, 2, 3\}$ a relation that is reflexive and symmetric but not transitive. 8 marks

1. **Reflexive** — $a - a = 0 = 3 \cdot 0$, so 3 divides $a - a$ and $a \; R \; a$. 2
2. **Symmetric** — if $a - b = 3m$ then $b - a = 3(-m)$, and $-m$ is an integer, so $b \; R \; a$. 2
3. **Transitive** — $a - b = 3m$ and $b - c = 3k$ give $a - c = 3(m + k)$, so $a \; R \; c$; hence R is an **equivalence relation**. 2
4. **Classes** by remainder on division by 3: $[0] = \{\ldots, 0, 3, 6, \ldots\}$, $[1] = \{\ldots, 1, 4, 7, \ldots\}$, $[2] = \{\ldots, 2, 5, 8, \ldots\}$ (all negatives included) — disjoint, union $\mathbb{Z}$. 2



inside one class every pair is related — that is transitivity inside a class

Counter-example On $\{1,2,3\}$ take $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$. Reflexive and symmetric, but **not** transitive: $(1,2)$ and $(2,3)$ would force $(1,3)$, which is absent.

Before the exam ✓ 2^n, $C(n,k)$ and $|A \times B|$ instantly · all set laws with symbols · the six properties on a small relation · equivalence vs partial order.